The Journal of Open Source Software

TCTrack: FAIR Tropical Cyclone Tracking

Jack W. Atkinson [1] **and Samuel J. Avis** [1¶]

1 Institute of Computing for Climate Science, University of Cambridge, UK ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Gabriele Bozzola

Reviewers:

- @1JunGu
- @leosaffin

Submitted: 24 April 2026
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

In partnership with

JOSS Special Issue for RSECon26 Research Software Papers

Summary

TCTrack is a Python package providing a unified, user-friendly interface to multiple established tropical cyclone tracking algorithms: TRACK (Hodges et al., 2017), TempestExtremes (Ullrich et al., 2021), and TSTORMS (Vitart et al., 1997). Whilst these algorithms are widely used in the climate science community, each has its own unique interface, input requirements, and output format, making intercomparison studies and routine application challenging. TCTrack addresses these usability issues by wrapping each algorithm with a consistent Python API based on abstract base classes. Crucially, TCTrack transforms the output from each algorithm into a standardised NetCDF format that is fully compliant with the Climate and Forecasting (CF) conventions (Earth Science Data Systems, 2024) using the discrete sampling geometry trajectory format, ensuring that all metadata from input data and processing parameters are preserved. This approach makes TCTrack particularly valuable for climate model evaluation studies, such as those in CMIP (the Coupled Model Intercomparison Project), and for tracking algorithm intercomparison work. By providing standardised, metadata-rich outputs, TCTrack aligns with the FAIR principles for research data and software (Barker et al., 2022; Wilkinson et al., 2016), ensuring outputs are Findable, Accessible, Interoperable, and Reusable.

Statement of need

Tropical cyclone tracking is an essential component of climate model evaluation and climate change impact assessment. However, researchers face significant barriers when attempting to use existing tracking codes. Each established algorithm—TRACK, TempestExtremes, and TSTORMS—requires separate installation, has distinct input data requirements and parameter specifications, and produces output in custom formats with limited or no metadata. Users who wish to compare results across multiple tracking algorithms must invest substantial effort learning each tool independently, converting between different data formats, and manually adding provenance metadata to outputs.

TCTrack streamlines this workflow by providing a uniform interface across all three algorithms. Users need only:

1) configure tracking parameters using algorithm-specific parameter dataclasses,
2) instantiate a tracker object and call the `run_tracker()` method,
3) use the CF-compliant netCDF output data.

All output files follow the CF conventions' discrete sampling geometry trajectory format (sections 9 and H4 of CF-1.13 (Earth Science Data Systems, 2024)), with coordinate variables, auxiliary coordinates, and complete metadata including algorithm name, TCTrack version, and all parameters required for reproducibility.

This standardised framework that TCTrack provides will support researchers, forecasters, and the natural hazard industry (e.g. insurers) for using different algorithms to project the impact of tropical cyclones. It is particularly valuable for two key use cases. First, climate model evaluation studies—such as the assessment of tropical cyclones in Neural GCM (Kochkov et

DRAFT

al., 2024)—require consistent application of tracking algorithms to model outputs. Second, intercomparison studies are essential given documented variations in cyclone statistics between different CMIP models (Bourdin et al., 2024), tracking methods (Roberts et al., 2020), and reanalysis datasets (Hodges et al., 2017). TCTrack facilitates such studies by providing a common framework whilst preserving the distinct characteristics of each underlying algorithm.

State of the Field

Three tracking algorithms provided by TCTrack (TRACK, TempestExtremes, and TSTORMS) are established, peer-reviewed methods used in the climate science community. TRACK uses a feature-tracking approach based on vorticity maxima and has been applied to assess tropical cyclones in reanalysis datasets (Hodges et al., 2017), TempestExtremes employs geometric criteria on multiple variables (Ullrich et al., 2021), and TSTORMS searches for co-located vorticity and temperature maxima and sea-level pressure minima (Vitart et al., 1997). However, their native interfaces are command-line based with algorithm-specific input formats and minimal output metadata, creating barriers to adoption and intercomparison.

Alternative Python-based tracking tools exist, such as tobac (Heikenfeld et al., 2019), which provides general feature tracking capabilities primarily for meteorological applications. However, tobac implements new tracking methodologies rather than providing access to the established algorithms that have been extensively used and validated in the climate community. TCTrack's approach differs fundamentally: it enables researchers to use proven methods whilst addressing the practical challenges of usability, interoperability, and metadata standards.

TCTrack is also extensible in that additional tracking algorithms can be integrated by following the contributor guidelines and building on the abstract base class design. Furthermore, despite the package being primarily focussed upon tropical cyclone tracking many of the existing tracking algorithms could also be used for extratropical cyclones. Indeed, provided an appropriate tracking algorithm, any sort of feature that could be represented by a time-based trajectory could be supported by TCTrack.

The decision to build TCTrack rather than change the data format/running procedures of multiple existing projects comes from the desire to also provide a unified interface for researchers as well as improved data representation. TCTrack's focus on usability and FAIR data builds on what the existing tools offer, making it a valuable part of the ecosystem.

Software Design

TCTrack is implemented as a Python package built using object-oriented design. Users select an appropriate parameter class to configure tracking, then pass this to a tracker class in order to run and output results. These use abstract base classes to present a standardised API across the different tracking algorithms and clear interfaces for adding new algorithms. Meanwhile, the algorithm-specific details are abstracted away from the end user by the concrete implementations of these classes.

It is possible for tracking algorithms to be implemented directly in TCTrack, but when building on existing tracking codes we utilise `subprocess` calls to reduce the software maintenance required and give users access to the proven underlying programs.

A particularly important aspect of TCTrack is the standardised, CF-compliant output format into which the output tracks are transformed; achieved using the cf-python software package (Hassell et al., 2017; Hassell & Bartholomew, 2020). This ensures that outputs are FAIR (which is of increasing importance in climate science) and immediately usable in downstream analysis, something that is not a feature of existing approaches. Metadata is added to the output by capturing the metadata from input files and augmenting it to reflect any processing

DRAFT

of variables, such as spatial aggregation. It also provides the tracking algorithm used, TCTrack version, and complete parameter specifications for reproducibility.

The codebase includes comprehensive unit tests using pytest and continuous integration via GitHub Actions to ensure validity and compatibility across multiple Python versions. Code quality is maintained through static analysis to provide robustness and maintainability.

The package is available via GitHub at https://github.com/Cambridge-ICCS/TCTrack and extensively documented at https://tctrack.readthedocs.io/. This includes installation guidance, API reference, and worked examples.

Research Impact Statement

TCTrack addresses a need within the community for simple, accessible tooling for exploring the differences in tropical cyclone tracks and statistics produced by different tracking algorithms and conducting intercomparison studies.

It has been used in climate science research as part of CMIP6 model comparison studies by researchers at the University of Cambridge, with some initial results presented to the scientific community in Atkinson et al. (2026). This work also aims to reproduce and update the findings of Knutson et al. (2020) to project the response of tropical cyclone activity to future climate change using more recent and higher resoultion CMIP6 data (paper in preparation). TCTrack is also being used in industry by INIGO Insurance to generate tracks in a robust, standardised format for downstream use in data analysis and presentation.

TCTrack has received collaboration from researchers and external users, who have opened feature requests and made direct code contributions. Notably, one researcher has integrated their new detection algorithm, based on a machine-learning classifier, in order to make use of TCTrack's software architecture and metadata pipeline.

Acknowledgments

This project is supported by a philantropic donation from INIGO Insurance as part of the InSPIRe project. We also thank the Institute of Computing for Climate Science for their support. We thank Alison Ming and Charles Powell for their testing and feedback during development.

AI Usage Disclosure

During the development of this software AI tools (ChatGPT, OpenCode / Claude Code agents running Devstral and GLM-5.3) were used to investigate and understand external software, plan and review changes to the code, and for the generation of tests. Any code suggestions by AI were thoroughly checked and double checked during code review by a second person. During the preparation of this paper, AI was used to refactor an initial draft to align with the changes to the JOSS scope. This was thoroughly checked and built upon thereafter.

References

Atkinson, J., Avis, S., Ming, A., & Powell, C. (2026). *TCTrack: Facilitating FAIR tropical cyclone tracking software and data*. https://doi.org/10.5194/egusphere-egu26-5825

Barker, M., Chue Hong, N. P., Katz, D. S., & others. (2022). Introducing the FAIR principles for research software. *Scientific Data*, *9*(1), 622. https://doi.org/10.1038/s41597-022-01710-x

DRAFT

Bourdin, S., Fromang, S., Caubel, A., Denvil, S., Mignot, J., Ladant, J.-B., Swingedouw, D., & Gastineau, G. (2024). Tropical cyclones in global high-resolution simulations using the IPSL model. *Climate Dynamics*, *62*(5), 4343–4368. https://doi.org/10.1007/s00382-024-07138-w

Earth Science Data Systems. (2024). *NetCDF climate and forecast (CF) metadata conventions* (Version 1.13) [Computer software]. https://cfconventions.org/Data/cf-conventions/cf-conventions-1.13/cf-conventions.html

Hassell, D., & Bartholomew, S. L. (2020). cfdm: A python reference implementation of the CF data model. *Journal of Open Source Software*, *5*(54), 2717. https://doi.org/10.21105/joss.02717

Hassell, D., Gregory, J., Blower, J., Lawrence, B. N., & Taylor, K. E. (2017). A data model of the climate and forecast metadata conventions (CF-1.6) with a software implementation (cf-python v2.1). *Geoscientific Model Development*, *10*(12), 4619–4646. https://doi.org/10.5194/gmd-10-4619-2017

Heikenfeld, M., Marinescu, P. J., Christensen, M., Watson-Parris, D., Senf, F., Heever, S. C. van den, & Stier, P. (2019). Tobac 1.2: Towards a flexible framework for tracking and analysis of clouds in diverse datasets. *Geoscientific Model Development*, *12*(11), 4551–4570. https://doi.org/10.5194/gmd-12-4551-2019

Hodges, K., Cobb, A., & Vidale, P. L. (2017). How well are tropical cyclones represented in reanalysis datasets? *Journal of Climate*, *30*(14), 5243–5264. https://doi.org/10.1175/JCLI-D-16-0557.1

Knutson, T., Camargo, S. J., Chan, J. C. L., Emanuel, K., Ho, C.-H., Kossin, J., Mohapatra, M., Satoh, M., Sugi, M., Walsh, K., & Wu, L. (2020). Tropical cyclones and climate change assessment: Part II: Projected response to anthropogenic warming. *Bulletin of the American Meteorological Society*, *101*(3), E303–E322. https://doi.org/10.1175/BAMS-D-18-0194.1

Kochkov, D., Yuval, J., Langmore, I., Norgaard, P., Smith, J., Mooers, G., Klower, M., Lottes, J., Rasp, S., Düben, P., & others. (2024). Neural general circulation models for weather and climate. *Nature*, *632*(8027), 1060–1066. https://doi.org/10.1038/s41586-024-07744-y

Roberts, M. J., Camp, J., Seddon, J., Vidale, P. L., Hodges, K., Vanniere, B., Mecking, J., Haarsma, R., Bellucci, A., Scoccimarro, E., & others. (2020). Impact of model resolution on tropical cyclone simulation using the HighResMIP–PRIMAVERA multimodel ensemble. *Journal of Climate*, *33*(7), 2557–2583. https://doi.org/10.1175/JCLI-D-19-0639.1

Ullrich, P. A., Zarzycki, C. M., McClenny, E. E., Pinheiro, M. C., Stansfield, A. M., & Reed, K. A. (2021). TempestExtremes v2.1: A community framework for feature detection, tracking, and analysis in large datasets. *Geoscientific Model Development*, *14*(8), 5023–5048. https://doi.org/10.5194/gmd-14-5023-2021

Vitart, F., Anderson, J. L., & Stern, W. F. (1997). Simulation of interannual variability of tropical storm frequency in an ensemble of GCM integrations. *Journal of Climate*, *10*(4), 745–760. https://doi.org/10.1175/1520-0442(1997)010%3C0745:SOIVOT%3E2.0.CO;2

Wilkinson, M. D., Dumontier, M., Aalbersberg, Ij. J., & others. (2016). The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, *3*(1), 160018. https://doi.org/10.1038/sdata.2016.18

DRAFT